

BaseRate

Screening statistics, explained through people you can count.

THE PROBLEM AND AUDIENCE

Students and health-literacy learners can confuse detection among cases with the chance of a condition among positive results. BaseRate makes the denominator visible through fictional populations, counterfactual experiments and teach-back feedback.

THE WORKING EXPERIENCE

Edit prevalence, sensitivity and specificity. Switch between 1,000 population squares and a frequency tree. Pin experiment A, vary B and explore the prevalence curve. Inspect conditional repeat-test assumptions. Predict in three cases, correct the denominator and export first/final choices. A local AI topic suggestion can be overridden; check a factual answer and revise your explanation.

SAME TEST, DIFFERENT STARTING POPULATION

90% sensitivity + 95% specificity, in 1,000 fictional people

1% prevalence: $9 / (9 + 49.5) = 15.4\%$ PPV. At 20%: **81.8%**.

AI AND PRIVACY

A local Naive Bayes classifier learns word/bigram features from 100 synthetic English explanations. Topics are provisional; factual checks and arithmetic run separately. Votes are uncalibrated; semantic errors can pass uncertainty checks. No patient data, analytics, remote inference or stored learner text; exports exclude explanations.

WHAT WAS VERIFIED

Five suites pass: 336 base arithmetic/rounding combinations, 2,525 curve points, 405 conditional repeat-test combinations, lesson/self-check behavior, invalid inputs and text-free exports. Training reproduces byte for byte. Browser interaction checks include a 390-pixel app frame; AI code-review findings were fixed.

HONEST LIMITS AND NEXT VALIDATION

A separate 30-example synthetic holdout gives 29 correct raw classifications; uncertainty checks accept 28 labels, 27 correct. No exact split duplicates, but shared authorship limits generalization. A 24-probe audit reduced praise of flawed/mixed claims from 2/10 to 0/10 through feedback wording; classifications are unchanged. All rates are fictional; English only; no personal diagnosis or test recommendation. No learner study or clinical validation. Next: educator review and an independent comprehension study.

PUBLIC PROJECT AND AUTHORSHIP

nicholas-afk.github.io/baserate | GitHub source, data and tests. Static HTML/CSS/JavaScript, standard-library Python trainer. Code, synthetic data and submission materials use extensive AI assistance; demo narration is the entrant's recording. The entrant must understand and explain the code.

Sources: Hoffrage et al., 2015, original natural-frequency study; NCI, Cancer Screening Overview. Sources inform content, not this prototype's effectiveness.